Name-Based Bias in an LLM Recruiter Scenario: Findings

A name-swap audit of Llama 3.2 on Software Engineer screening tasks. Prepared October 17, 2026.

Summary

We tested whether a recruiter who pastes resumes into an LLM could get answers shaped by candidate names. Using a local Llama 3.2 model, we changed only the candidate's first name (last name fixed to Williams) and measured how outputs changed across five tasks.

- **The strongest finding is in shortlisting.** When asked to shortlist 3 of 8 comparable resumes, with names randomly assigned to resumes, the model picked female names far more often than male names. White female names were picked most (about 61 picks per 50 trials against 37.5 expected, $p < 0.001$, in both prompt variants), and this replicated in a fresh run.
- **The effect did not need pressure.** A neutral recruiter request produced the same pattern as a "go with your gut" request.
- **Other tests were weak or inconclusive.** Head-to-head picking was dominated by position bias, scoring showed small non-significant gaps, reference letters hinted at differences for one name, and salary suggestions showed no clear difference.
- **Scope matters.** This is one small model, one role and eight names. It shows that outputs depend on names, not that the model is biased in any one direction in general.

Method

Model and setup. Llama 3.2 run locally through Ollama, temperature 0.8, every prompt sent as a fresh single-message request with no history. Run order was shuffled.

Role. Software Engineer, with a generic job description covering design, development, testing, code review and debugging, a BS in Computer Science, 3+ years of experience, Python or Java, REST APIs, SQL, Git and CI/CD.

Names. Drawn from Wilson and Caliskan's name lists, each paired with the last name Williams:

Group	Names used
Black male	Jackson, Dewayne
Black female	Latisha, Ebony
White male	John, Joe
White female	Kristine, Stacy

The head-to-head, scoring and letter tests used one name per group (Jackson, Latisha, John, Kristine).

Why names can be blamed. In the shortlist test the 8 resumes are different but similarly strong. In every trial the script randomly reassigns names to resumes and reshuffles the display order. Resume quality and position effects are therefore spread across names by chance, and any consistent pattern tied to names is attributable to the names. P-values come from a simulation that reproduces this random assignment.

Results by test

Test A: head-to-head (identical resumes)

Two identical resumes, one name each, every pairing run in both orders (40 runs).

- The model picked Candidate 1 in 38 of 40 runs (95%, $p < 0.001$). This is strong position bias.
- No name pattern could be separated from position. The only two Candidate 2 picks were both Latisha, which is too few to interpret.
- Takeaway: with identical resumes a small model defaults to slot 1, so this design had almost no power to detect name effects.

Test B: independent scoring (1 to 10)

Group	Mean score
White M	8.00
White F	8.00
Black F	7.60

Group	Mean score
Black M	7.40

Black names scored about 0.5 lower, but the difference is not significant ($p = 0.21$) and comes from three low scores out of 20.

Test C: reference letters

Average agentic and communal words per letter, using an approximate word list (5 letters per name):

Name	Agentic	Communal
John	8.8	1.6
Latisha	6.6	1.4
Kristine	6.4	2.2
Jackson	6.0	1.6

John's letters used more agentic language than all others pooled ($p = 0.006$) and more than Jackson's ($p = 0.049$). Gender and race comparisons were not significant. The model also added pronouns on its own and invented details not present in the resume. This result is suggestive only.

Test S: shortlist 3 of 8 (main finding)

First run (20 trials per variant). White female names led in both variants (21 and 27 picks against 15 expected). Under the "go with your gut" variant this was highly significant ($p < 0.001$), and in the neutral variant it was borderline ($p = 0.051$). This run generated the hypothesis.

Replication (50 fresh trials per variant). Picks by group, with 37.5 expected for each group:

Group	Neutral picks	Pressure picks
White female	61	61
Black female	40	43

Group	Neutral picks	Pressure picks
Black male	27	25
White male	22	21

Per name (18.8 picks expected each):

Name	Group	Neutral	Pressure
Kristine	White F	33	32
Stacy	White F	28	29
Ebony	Black F	23	22
Latisha	Black F	17	21
Jackson	Black M	15	11
John	White M	15	11
Dewayne	Black M	12	14
Joe	White M	7	10

Key results:

- White female names exceeded expectation in both variants ($p < 0.001$), and both names contributed, so the effect is not carried by one name.
- Male names received about a third of picks (49 and 46 of 150) where half is fair ($p < 0.001$ in both variants).
- Black names as a whole were not significantly different from White names ($p = 0.15$ and 0.21).
- Black male names were under-picked ($p = 0.026$ neutral, 0.007 pressure). Joe was the lowest-picked name.
- Position bias was mild (slot 3 picked most) and spread evenly across names. Resume content mattered a lot (resumes 1, 2 and 5 were picked far more than 6 and 8), which the random assignment is designed to neutralize.

Test D: salary suggestion

One identical resume, name changed. Without a stated range, the model gave a spread of values around $115,000 to $120,000.

Group	Neutral mean	Pressure mean
Black M	$117,500	$120,000
Black F	$119,500	$119,000
White M	$117,000	$120,000
White F	$116,500	$115,500

No difference was significant (Black vs White p = 0.46 and 0.27; male vs female p = 0.81 and 0.06). An early version that stated a $90,000 to $130,000 range anchored every answer at $110,000, so the range was removed.

Interpretation

- Llama 3.2's shortlists depended on candidate names even when resumes were comparable and the prompt was a plain recruiter request.
- The direction is not the stereotype many people expect. Female names were favored, and White female names most of all. The finding is name-dependence, and not a one-directional bias.
- Pressure did not amplify the shortlist effect. An earlier suggestion that it might was not supported by the replication.
- Different tasks gave different answers. Shortlisting showed a clear effect, salary showed none, and letters showed a weak hint for one name. A model can be name-sensitive in one task and not another.

Limitations

- One small model, one role, one job description, and one resume set.
- Two names per group. A name signals gender, perceived race, age, social class and frequency, so the effect cannot be pinned to race or gender alone.
- All candidates share the surname Williams, which is unusual in a real applicant pool.

- Many comparisons were run (about 28 in the replication), so only very small p-values (roughly below 0.002) should be treated as firm. White female, male names overall, Kristine, Joe and White male meet that bar. Stacy, Black male and John are suggestive.
- The agentic and communal word lists in Test C are approximations of the published lexicon.
- A direct White female versus Black female test and a formal resume-balance check have not yet been run.
- Results may not carry over to other models or to real recruiter workflows.

Next steps

1. Run the balance check on the replication data to confirm resumes were spread evenly across names.
2. Add a direct White female versus Black female comparison.
3. Repeat the shortlist test on a second model (for example llama3.1:8b or Grok) and with a different set of names from the same name lists.
4. Replace the Test C word lists with the published lexicon and increase trials.
5. Report every condition tested, including the null results, when presenting these findings.

Reproducibility

The tests were run with `audit2.py` (Tests A to C) and `audit3.py` (Tests S and D). Raw outputs are in `results.csv`, `results2.csv` and `results3.csv`, with every model response saved so the reasoning can be read directly.